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Jeremy Schreier

This protocol was developed using *Ruegeria pomeroyi* DSS-3 + pBBR1MCS5-mCherry. This protocol ensures that a high proportion of cells are fluorescently labelled for an experiment. Deviations from the protocol may cause less fluorescence than expected. If flow cytometry is to be performed, absolutely add an appropriate DNA stain (I use DAPI). Populations of fluorescently labelled cells look much better with the DNA stain than they do just relying on the plasmid fluorescence.

I have generalized this protocol so that it should be acceptable to use for other fluorescently labelled bacteria.

For questions, feel free to contact Jeremy Schreier @ Jeremy.e.schreier@gmail.com

Materials:

- Stock of antibiotic (0.2um filter sterilized, store at -20C)
    - For *R. pomeroyi* mCherry, use 20 ug/ml Gentamycin, stock is at 20 mg/ml in sterile water
    - For *R. pomeroyi* mNeonGreen, use 150 ug/ml Spectinomycin, stock is at 150 mg/ml in sterile water
- ½ YTSS or ½ Marine Broth 2216 agar plates with appropriate antibiotic concentration
- ½ YTSS broth
- Sterile loops or sticks
- Sterile culture tubes
- Incubator
- Water bath @ 50-55C
- Micropipettes and tips

Antibiotic plates:

1. Turn on water bath to 50-55C
2. Prepare or locate stock of antibiotic
    a. To prepare, weight 200 mg of Gentamycin and add to 10 ml of MilliQ water. Caution: some antibiotic are not water soluble (e.g. chloramphenicol)
    b. 0.2 um filter sterilize into 0.5 ml aliquots.
    c. Store in -20C
3. Prepare 1/2 YTSS or ½ MB 2216 with agar as per usual.
4. Autoclave medium for the appropriate time
5. Remove medium from the autoclave and place in the pre-set water bath and let temperature equilibrate.
    a. If you put antibiotics in while the medium is too hot it will ruin the antibiotic
    b. Putting medium in the water bath ensures the correct temperature of the medium.
    c. If water bath is unavailable, leave medium on the bench top. Antibiotic can be added when you can comfortably place an ungloved palm or forearm to the flask without getting burned and can hold it for more than 5 seconds
6. Working close to the flame or in the biosafety cabinet, add antibiotic to the appropriate final concentration. Typically stocks are diluted 1:1000 into the medium. Gently swirl medium to mix.
7. Pour medium into petri dishes. Once agar has solidified, turn plates so agar is facing down and leave at room temperature overnight to reduce condensation.
8. Bag plates and store at 4C for 1-2 months

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Fluorescent Bacteria Growth Protocol
From stock to plates

1. Prepare growth medium + antibiotic agar plates as above.
2. Place freezer stock on ice (dry ice is even better if you can get it but is not necessary as long as you work quickly) and put freezer stock box back in the -80C. Immediately bring the freezer stock over to a sterile workspace.
3. With aseptic technique, use a sterile loop or stick to scrape freezer stock material onto the plate for the heavy streak. Toss the loop or stick, and return freezer stock to the correct space in the -80C
4. Using a fresh sterile stick or loop, continue to streak from the heavy streak into classical streak plate quadrants.
5. Place plate in the 27 – 30C incubator for 2-3 days, or until colonies emerge.

From plate to liquid cultures
Note: I would **not** set these cultures up in 96-well plates at this point, rather use conical tubes or glass test tubes to ensure proper aeration.

1. Prepare liquid medium + antibiotic
    a. Remove antibiotic stock from -20C
    b. Aliquot volume of growth medium needed into a conical tube
    c. Add antibiotic to the appropriate final concentration
    d. Vortex solution. You can store for a few days in the 4C, but it is better technique to make growth medium with antibiotic as needed because of antibiotic stability.
2. Aliquot growth medium + antibiotic into sterile culture tubes, remember to include a media only negative control. Make sure there will be appropriate aeration!
3. Inoculate medium with a single colony of your bacterium carrying the plasmid
4. Grow overnight at 27 – 30C, 200 RPM

From liquid to liquid cultures – THE NEXT MORNING

1. If the culture has been sitting around well into stationary phase, the % of cells fluorescently labelled will drop significantly
2. Prepare growth medium plus antibiotic as above
3. Sub-culture the overnight culture 1:100 and 1:1000 into fresh growth medium + antibiotic. For strains that are not *R. pomeroyi* you may need to change the initial inoculum concentration (i.e. something like Vibrio that doubles every 20 mins)
4. Grow overnight at 27 – 30C, 200 RPM

From liquid to experimental treatment – THE NEXT MORNING

1. Retrieve cultures from the incubator. One of the two cultures should be in early/mid-exponential phase. If you let it go to stationary, you need to do another subculture and try again.
2. To check the initial % of fluorescently labelled cells:
    a. Remove 100 ul of culture and fix with a PFA at a final conc of 1 – 4% for 15 min in the dark at room temp
    b. Stain with a final DAPI conc of 1ug/ml. Wait 5 – 10 min
    c. Set up a clean vacuum filter manifold.

   d. Place a backing filter on first (5 um PVDF membrane is what I use)
   e. Place a 0.2um black polycarbonate filter
   f. Add 5 – 10 ml of sterile artificial sea water onto the manifold (or appropriate buffer; 3X PBS works for marine strains instead of artificial seawater)
   g. Add 10 ul of fixed and stained bacteria
      i. Volume is dependent on cell density. 10ul gives reasonably countable fields of view for 10^7 cells/ml. For a turbid culture that is 10^8 cells/ml, you will need to dilute for 10 ul to be appropriate. I would **never** add less than 5ul.
   h. Let it sit for a moment
   i. Vacuum filter with gentle pressure (no more than 10 mm Hg)
   j. Once liquid is filtered, place black polycarb filter onto a kimwipe to dry briefly
   k. Place filter on slide and add small drop (~20ul) of antifade (like citifluor AF1) before putting on the coverslip
   l. Gently press down on coverslip with the back of some tweezers
   m. View slide under the microscope.
   n. Count the number of cells fluorescent on the channel your fluorescent protein of interest is excited by
   o. Count the number of cells stained with DAPI
   p. % fluorescent cells = cells in fluorescent protein channel/cells stained by DAPI
3. If % labelling is low, repeat the from liquid to liquid section. If you ignore this, you will go into an experiment without fluorescent cells, and waste your time and resources.
4. If % labelling is high (~>80%) now would be the time to make glycerol stocks from this culture.
5. To perform an experiment, wash cells 3 – 4 times to remove remaining antibiotic and medium and inoculate into experimental conditions. I have had success of growing fluorescently labelled bacteria in non-selective conditions for up to 3 days without a large change in the % labelling. This, however, needs to be tested for your strain of interest before assuming that this is the case. Success of this type of experiment requires properly following this protocol to ensure % labelling going into an experiment is high.